MIL-T-81784C(AS)
11 July 1977
Superseding
MIL-T-61734B(AS)
6 December 1974

# MILITARY SPECIFICATION
# TARGET SIMULATOR, INFRARED
# SM-630( )/AWN-23(V)

This specification has been approved by the Naval Air Systems Command, Department of the Navy

## 1. SCOPE

1.1 Scope This specification covers the performance, design, and test requirements for the Infrared Target simulator (IRTS) SM-630( )/AWM-23(V).

## 2. APPLICABLE DOCUMENTS

2.1 General. The following documents, of the issue specified in AS-4055, Applicable Document List for AWCS Test Set AN/AWM-23(V), form a part of this specification to the extent specified herein MIL-T-81734(AS) shall be the applicable general specification for the IRTS described herein (see 6.6).

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-M-7793 | Meter, Time Totalizing |
| MIL-T-21200 | Test Equipment for Use with Electronic and Fire Control Systems, General Specification for |
| MIL-T-81734(AS) | Test Set, Airborne Weapon Control System AN/AWM-23(V) |

Naval Air Systems Command

| | |
|---|---|
| AS-2198 | Detail Specification, Infrared Subsystem of F-14A Weapon Control System (AN/AWG-9) |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to.

Engineering Specifications and Standards Department (Code 93) Naval Air Engineering Center, Lakehurst, NJ 08733, by using self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.



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STANDARDS

Military

| | |
|---|---|
| MIL-STD-461 | Electromagnetic Interference Characteristics Requirements for Equipment |

PUBLICATIONS

Hughes Aircraft Company

| | |
|---|---|
| 3282210 | Acceptance Test Procedure, Infrared Target Simulator |

2.2 Availability of documents. Applicable (see 6.6)

2,3 Precedence of documents. When the requirements of the contract, this specification, or applicable subsidiary documents are in conflict, the following precedence shall apply:

a. Contract. The contract shall have precedence over any other document.
b. This specification. This specification shall have precedence over all applicable subsidiary documents (see 6.6)
c. Referenced documents. Any document referenced in this specification shall have precedence over all applicable documents referenced therein.

3. REQUIREMENTS

3.1 Materials, parts, and processes. Applicable (see 6.6).

3.2 Design and construction. Applicable.

3.2.1 Design features. Applicable.

3.2.1.1 Method of operation. Applicable.

3.2.1.2 Commercial test equipment. Not applicable.

3.2. 1.2.1 Other commercial components. Applicable.

3.2.1.3 Circuit grounds. Applicable.

3.2.1.4 Interchangeability. Applicable.

3.2.1.5 Identification of product. Applicable.

3.2.1.6 Human factors. Applicable.

3.2.1.7 Test set generated acoustic noise. Applicable.



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3.2.1.8 Forced liquid cooling. Not applicable

3.2.1.9 Space and weight limitations. The maximum dimensions of the IRTS shall be 24 inchcs by 14 inches by 14 inches. The maximum weight shall be 50 pounds

3.2.1.10 Connectors. Applicable

3.2.1.11 Secondary electrical power. Not applicable.

5.2.1.12 Interface The simulator shall interface with the Receiver, Infrared R-1786( )/AWG-9.

3.2.2 Utilities. Applicable.

3.2.2.1 Electrical power. Applicable. The IRTS shall require not more than 300 volt-amperes of 115-volt, 400 Hertz (Hz), single-phase, Type I power at a 0.9 power factor.

3.2.2.2 Clean air Not applicable

3.2.2.3 Potable water-. Not applicable.

3.2.2.4 Liquid coolant. Not applicable.

3.2.2.5 Helium Not applicable

3.2.2.6 Nitrogen. Not applicable

3.2.2.7 Hydraulic fluid, Not applicable.

3.2.3 Service conditions

5.2.3.1 Electrical Applicable

3.2.3.2 Environmental. Applicable.

3. 2.3.2.1 Organizational level. Applicable, except that the operating temperature range shall be from -10 to +55°C. The +71°C intermittent operating temperature extreme of MIL-T-21200 shall not apply.

3.2. 3.2.2 Intermediate level. Applicable, except that the operating temperature range shall be from -10 to 55 C. The +71 intermittent operating temperature extreme of MIL-T-21200 shall not apply.

3.2.4 Maintainability. Applicable.

3.2.4.1 Mean time to repair Applicable.

3.2.4.2 Test points. Applicable.



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3.2.4.3 Shop replaceable assemblies (SRAs). Applicable.

3.2.4.4 Self test capability. Not applicable.

3.2.5 Reliability. The mean-time-between-failures (MTBF) shall be 500 hours minimum.

3,2.5.1 Operating life. Applicable.

3.2.5.2 Storage life. Applicable.

3.2.5.3 Operational stability. Applicable.

3.2.6 Electromagnetic interference (EMI). Applicable except as follows:

a. MIL-STD-461, Tests CE01 and CE02 shall not apply.
b. MIL-STD-461, Tests CE03 and CE04 and the limits therein shall apply except that transients occurring during functional non-measurement periods may exceed the limits by 40 db as measured at the simulator connector interface.
c. MIL-STD-461, Tests RS01, RS02, and RS04 shall not apply.

3.2.7 Utilitles protection. Applicable.

3.2.7.1 Liquid cooling protection. Not applicable,

3.2.7.2 Clean air protection, Not applicable.

3.2.7.3 Electrical power protection. Applicable.

3.2.8 Calibration cycle. Applicable.

3.3 Performance.

3.3.1 General, The simulator shall provide three calibrated irradiance simulated infrared (IR) targets to the infrared cryogenic refrigerator receiver. The simulated IR targets shall be selectable by the remote control specified in 3.6.8,

3.3.2 Organizational level functional requirements. Applicable,

3.3.3 Intermediate level functional requirements. Applicable when used with the Low Frequency Test Station, OJ-215( )/AWM-23(V).

3.3.4 Warmup time. Applicable.



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3.4 Detailed functional requirments The simulator shall provide calibrated irradiance equivalent to that provided from an 800 Kelvin blackbody. The lowest level of equivalent effective irradiance (EER) shall be 1.25 times the minimum infrared subsystem track mode irradiance specified in AS-2198 to an accuracy of +20 percent. Two additional effective irradiance levels shall also be provided which shall be 2 + 10 percent and 10 + 10 percent times the lowest level of EER specified above.

3.5 Workmanship Applicable.

3.6 Elements. The simulator shall include an optical collimator, blackbody and temperature controller assembly, digi-motor, aperture discs, and neutral density filters as elements in the functional design.

3.6.1 Optical collimator. The optical collimator shall collimate the IR radiation from the blackbody radiator into a one inch parallel light beam.

3.6.2 Blackbody and temperature controller The blackbody and temperature controller shall provide an IR target source with the following characteristics

| | | |
|---|---|---|
| a. | Temperature | 800 $+2^{O}$ Kelvin , measured at the apex-of the cavity |
| b. | Warmup time, standard conditions | 5 minutes, maximum |
| c. | Warmup time, extreme temperature conditions | 20 minutes, maximum |

3.6.3 Digi-motor. A controllable, unidirectional motorized stepping switch shall be provided to control the position of neutral density filters and aperture discs between the IR source and the optical collimator. The digi-motor shall have the following characteristics.

| | | |
|---|---|---|
| a. | Stepping increments | 30 +2 degrees |
| b. | Stepping rate | 3 steps per second |

3.6,4 Aperture discs. Four aperture discs, positioned by the digi-motor shall be provided. Three discs shall control the effective IR radiation level by means of varying apertures and the fourth disc shall completely interrupt the radiation. Apertures shall be as follows:

| | | |
|---|---|---|
| a. | Disc No. 1 | 0.0550 $\pm$0.0005 inch |
| b. | Disc No. 2 | 0 0580 $\pm$0.0005 inch |
| c. | Disc No. 3 | 0.0610 $\pm$0.0005 inch |
| d. | Disc No. 4 | No aperture |

3.6.5 Neutral density filters. Three neutral density filters shall be provided to control the IR spectrum characteristics. The filters shall be positioned by the digi-motor and shall be synchronized with the corresponding aperture disc.



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3.6.6 Elapsed time meter. An elapsed time meter, conforming to MIL-M-7793, shall be provided. The meter shall be readily visible during operation of the simulator and shall be capable of registering 9,999 hours.

3.6.7 Power cables. A detachable electrical power cable shall be provided in accordance with MIL-T-21200. The connector shall be MS3107A-16S-8P with MS3057-8A cable clamp. The connector shall have 115 volt, 400 Hz, single phase power on Pin A and return on Pin D. The length of the cable shall be 30 + 1.0 feet.

3.6.8 Control cable. A detachable remote control cable with an irradiance level selector switch shall be provided in accordance with MIL-T-21200. The length of the cable shall be 30 + 1.0 feet.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Applicable.

4.2 General. Applicable.

4.2.1 Classification. Applicable.

4.2.2 Individual items. Applicable.

4.2.3 In-process inspection. Applicable.

4.2.4 In-process data, Applicable.

4.2.5 Acceptance test plan. Applicable. Table I shall apply.

4.2.6 Test results. Applicable.

4.2.7 Acceptance. Applicable.

4.2,8 Utilities. Applicable.

4.2.9 Configuration. Applicable.

4.3 Tests.

4.3.1 Electrical and electronic items - operational run-in test. Applicable.

4.3.2 Mechanical items. Applicable.

4,4 Environmental test conditions. Applicable.

4.5 Test location. Applicable.

4.6 Acceptance schedule. Applicable.



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TABLE I

ACCEPTANCE TEST PLAN FOR INFRARED TARGET SIMULATOR

SM-630( )/AWM-23(V)

| | |
|---|---|
| 1.0 | TEST OBJECTIVE. To demonstrate functional performance of the Infrared Target Simulator. |
| 2.0 | TEST CONDITIONS |
| | a. Test Equipment |
| | (1) IRTS Test Position (HAC Part No. T493505-61-1) |
| | b. Utilities Required (Information only, ref. 4.2.8) |
| | (1) Power in accordance with 3.2.2 1 |
| 3.0 | TESTS. In accordance with Acceptance Test Procedure 3282210. |
| | a. Operational run-in-tests are not required. |
| | b. Measure the effective output irradiance of the IR target simulator in accordance with the acceptance test procedures. |

4.7 Test procedures. Applicable.

4.8 Rejection and retest. Applicable

5. PREPARATION FOR DELIVERY

5.1 General. Applicable.

6. NOTES

6.1 Intended use. The simulator is intended for use at the organizational and intermediate levels to facilitate checkout and test of certain weapon replaceable assembles (WRAs) of the infrared subsystem of Weapon Control System (WCS) AN/AWG-9.

6.2 Ordering data. Applicable.

6.3 Definitions. Applicable.



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6.4 Nomenclature cross-reference, The following is a cross reference between the official Navy nomenclature and the major contractor part numbers. The contractor part number suffix is controlled by the configuration list in the contract.

| Official Nomenclature | Contractor Part No. |
|---|---|
| a. Receiver, Infrared R-1786/AWG-9 | 481101-1XX |
| b. Target Simulator, Infrared SM-630 ( )/AWM-23(V) | 493505-1XX |
| c. Test Station, Low Frequency OJ-215( )/AWM-23(V) | 493350-1XX |

6.5 Liquid cooler. Not applicable.

6.6 Use of this specification. This specification is a detail specification which uses MIL-T-81734(AS) as a general specification. Where the corresponding paragraphs of MIL-T-81784(AS) are the same as MIL-T-81734(AS) the word “applicable” is used in this specification, When the term “not applicable” is used in this specification, it means that the corresponding paragraph requirements of MIL-T-81734(AS) are not required to be met (see 6.6 of MIL-T-81734(AS)).

6.7 Abbreviations and acronyms. Applicable. In addition the following shall apply.

| | |
|---|---|
| EER | Equivalent effective radiance |
| IR | Infrared |
| IRTS | Infrared target simulator |

6.8 Convenience outlet power. Not applicable.

6.9 Changes from previous issue are not marked.

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DOCUMENT IDENTIFIER AND TITLE

MIL-T-81784C(AS) Target Simulator, Infrared SM-630( )/AWM-23(V)

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